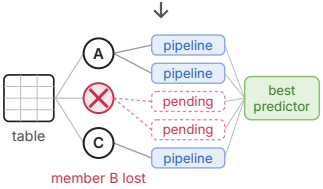
1 A team loses a member

A public science table: forest cover type, MiniBooNE neutrino events or HIGGS collider events



Three agents build classifier pipelines (prepare, fit, predict, validate). Halfway, one is shut down and its model withdrawn (fault F1). 18 cases: 3 tables × 2 workflow templates × 3 seeds.

2 Recover three ways

Fixed takeover. The next model in line redoes the missing work.

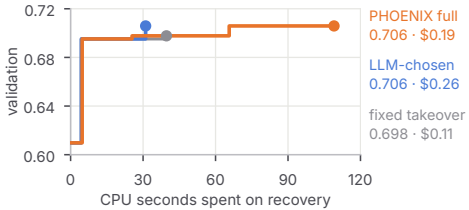
LLM-chosen. A planner model picks each step: keep, verify, recompute, reroute or stop.

PHOENIX full. The planner proposes plans; a search prices each in dollars and CPU along the dependencies, reuses valid results, and runs the plan with the most expected gain per unit of budget.

Equal caps per recovery: $0.20 (main study) or $1.00 (arms) of API, 8 model calls, 24 tool runs, 600 CPU s. **Planner:** GPT-6 Luna (main); Claude Opus 5.5 or GPT-6.1 Sol (arms, two draws per case).

3 Validation, then a sealed test

While recovering: one HIGGS case, Opus planning



Sealed test, scored once: 330 recoveries, paired by case

